TunnelForge Project Gantt Chart

College-project implementation plan | 28 September - 6 November 2026

PROJECT GOAL	SCOPE	SUCCESS CONDITION
Expose a localhost HTTP application through a relay and persistent WebSocket.	Core tunnel, basic authentication, reliability, tests, Docker deployment and demonstration.	A public request reaches localhost and its complete HTTP response returns to the caller.

Schedule note: Week 3 is highlighted in amber because 12 October 2026 is the current reporting date. Blue bars show planned activity.

Detailed Weekly Timeline

Each week ends with a clear, demonstrable milestone.

Week 1

28 Sep-2 Oct

Plan and design

Freeze the college-project scope; draw the architecture and request sequence; create the Go module and package structure; define versioned JSON messages for registration, requests, responses, heartbeat and errors.

Milestone: Architecture and protocol ready.

Week 2

5-9 Oct

Connect CLI and relay

Build the relay HTTP server and WebSocket endpoint; create an in-memory tunnel registry; generate secure tunnel IDs; implement CLI arguments and shared-token registration; return and print the temporary URL.

Milestone: CLI registers and receives a URL.

Week 3

12-16 Oct

Complete the first tunnel flow

Accept a public HTTP request; resolve its tunnel; send method, path, query, headers and body to the CLI; call the localhost application; return status, headers and body to the original caller.

Milestone: One complete end-to-end request works.

Week 4

19-23 Oct

Reliability and concurrency

Add request IDs and response matching; support concurrent requests; add body and queue limits; implement authentication checks, timeouts, heartbeat handling, reconnect behavior and meaningful errors.

Milestone: Stable tunnel under concurrent use.

Week 5

26-30 Oct

Test and deploy

Complete unit and integration tests; test GET/POST, JSON, headers, binary data, failures and concurrency; package the relay with Docker; add health checks and deploy to a public container host.

Milestone: Tested public relay deployment.

Week 6

2-6 Nov

Finalize and present

Fix defects; complete README, setup guide, diagrams and limitations; record results; prepare slides; rehearse a live demo and keep a recorded backup demonstration.

Milestone: Submission-ready project.